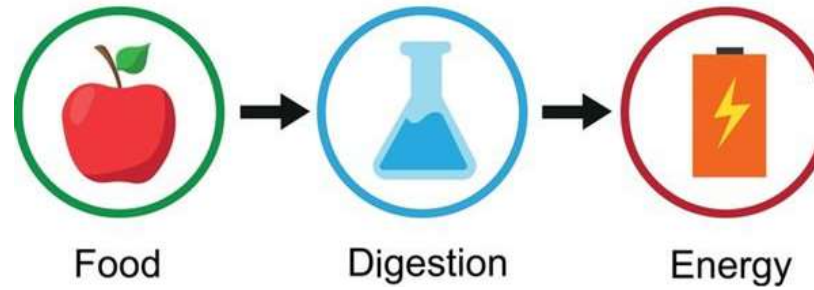


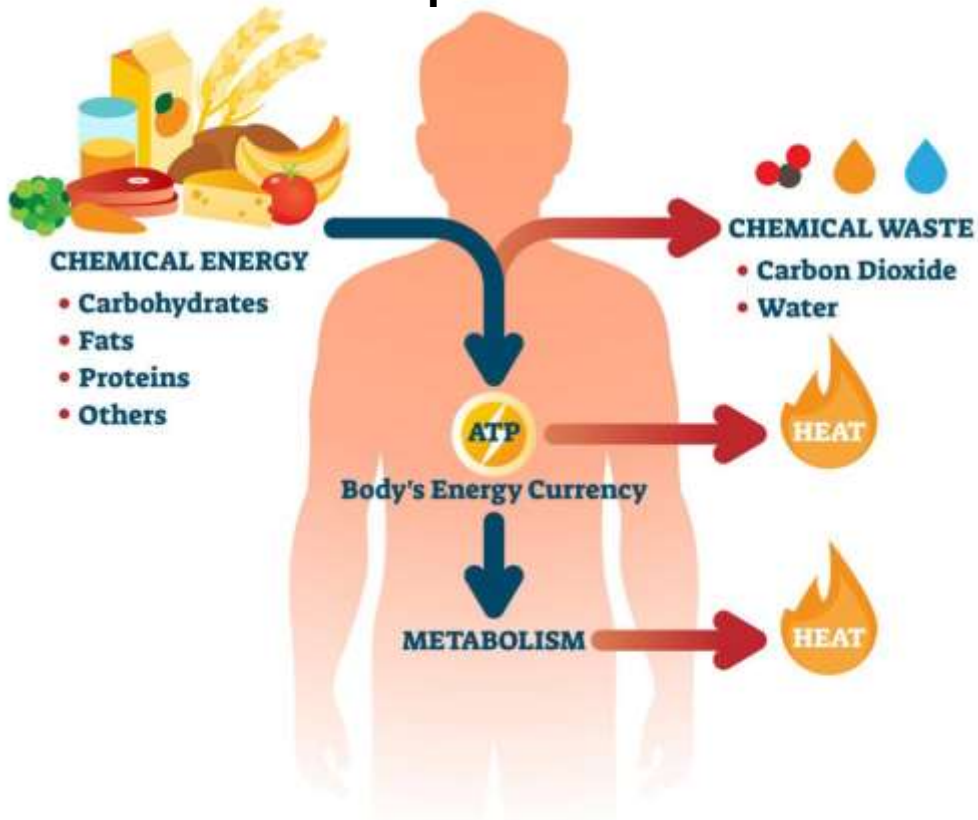
Photosynthesis and Cellular Respiration in Plants and Animals



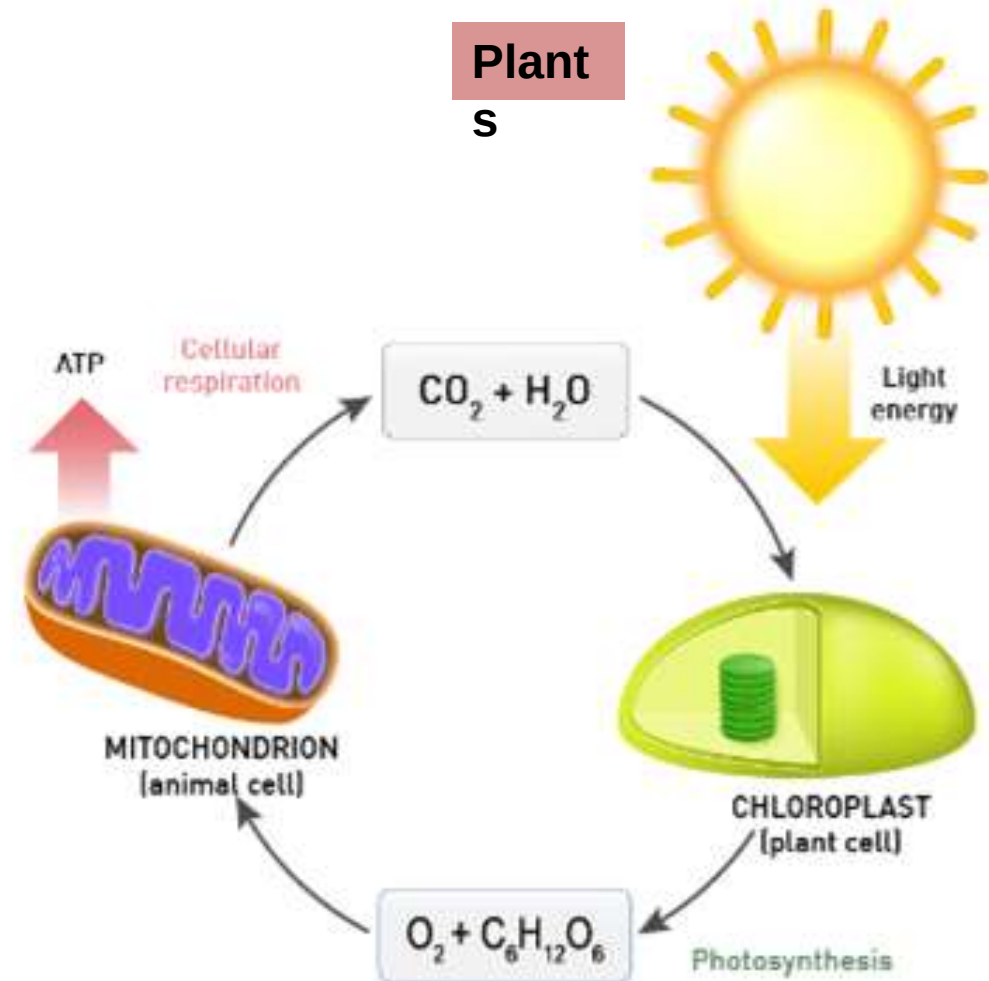
Metabolism



Animals



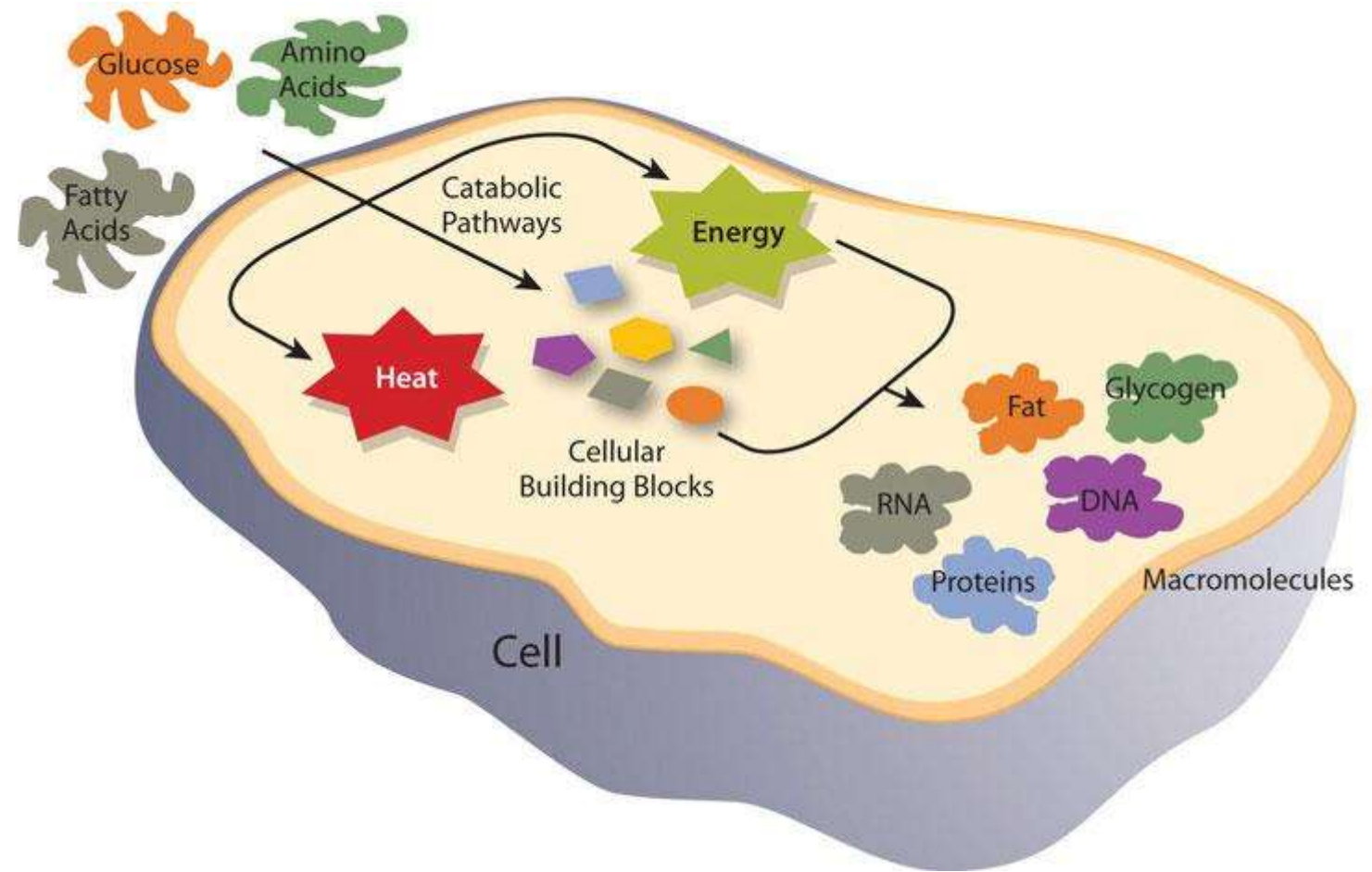
Plants



Metabolism

Metabolism is collectively **all the chemical reactions** that take place within each cell **of a living organism** for,

- providing energy to vital processes
- for synthesizing new organic material.
- Cells store energy by building organic molecules and retrieve energy by breaking them down.



Metabolic pathways

Anabolic: Small molecules are assembled into large ones. *Energy is required.*

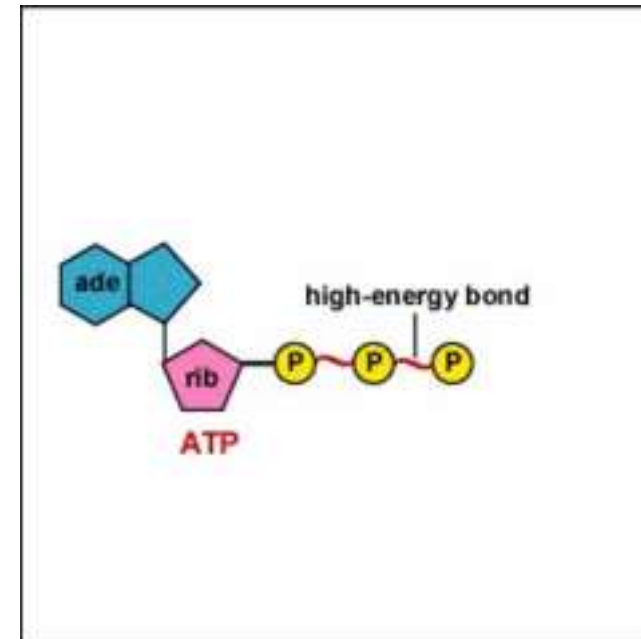
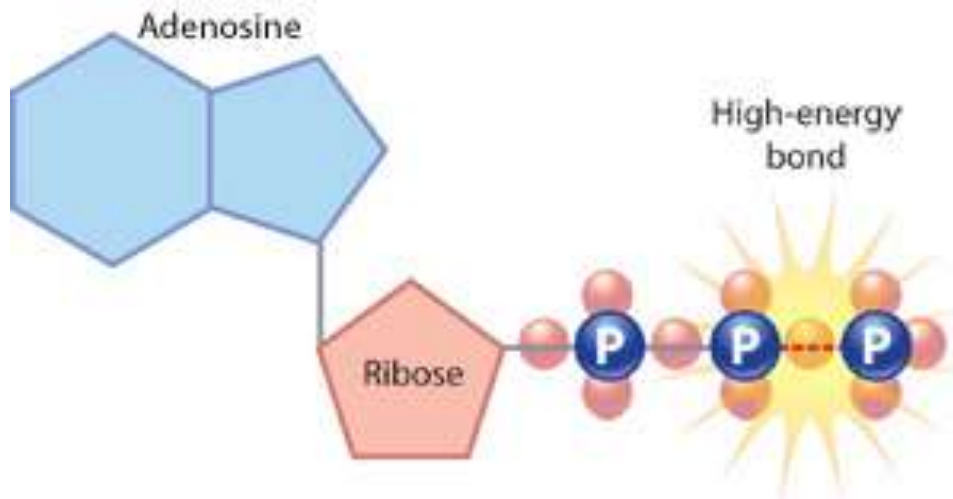


Catabolic: Large molecules are broken down into small ones. *Energy is released.*



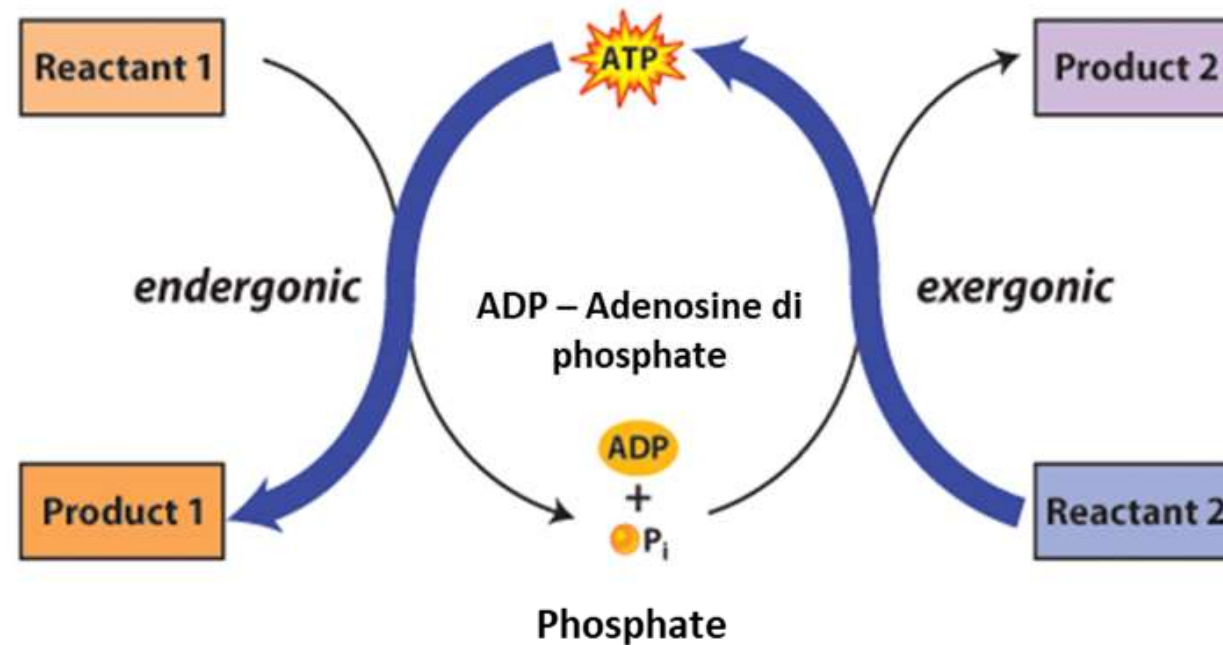
What is (ATP)?

- Adenosine triphosphate
- ATP is the energy-carrying molecule found in cells of living organisms.
- It is composed of :
 - One adenosine (adenine + ribose sugar) & three phosphate groups
 - One adenosine + two phosphate=ADP (Adenosine diphosphate)



The Role of ATP?

- Cells “earn” energy (ATP) in exergonic reactions
- Cells “spend” energy (ATP) in endergonic reactions



Photosynthesis and Cellular Respiration

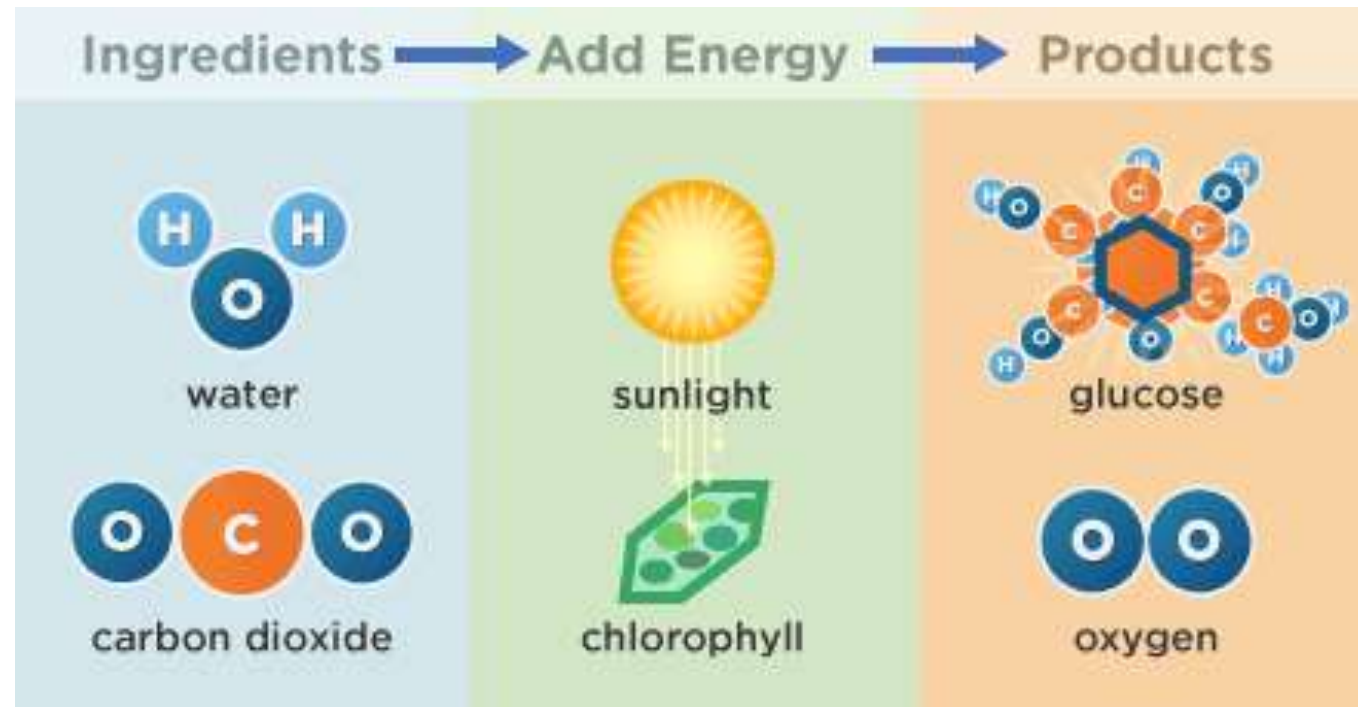
Photosynthesis and cellular respiration are two fundamental biological processes that sustain life on Earth.

While photosynthesis converts light energy into chemical energy, cellular respiration releases energy from organic molecules to fuel cellular activities.

These processes are interconnected and essential for the survival of plants and animals.

Photosynthesis

- Photosynthesis is the process by which green plants, algae, and some bacteria convert light energy into chemical energy, storing it in the form of glucose.



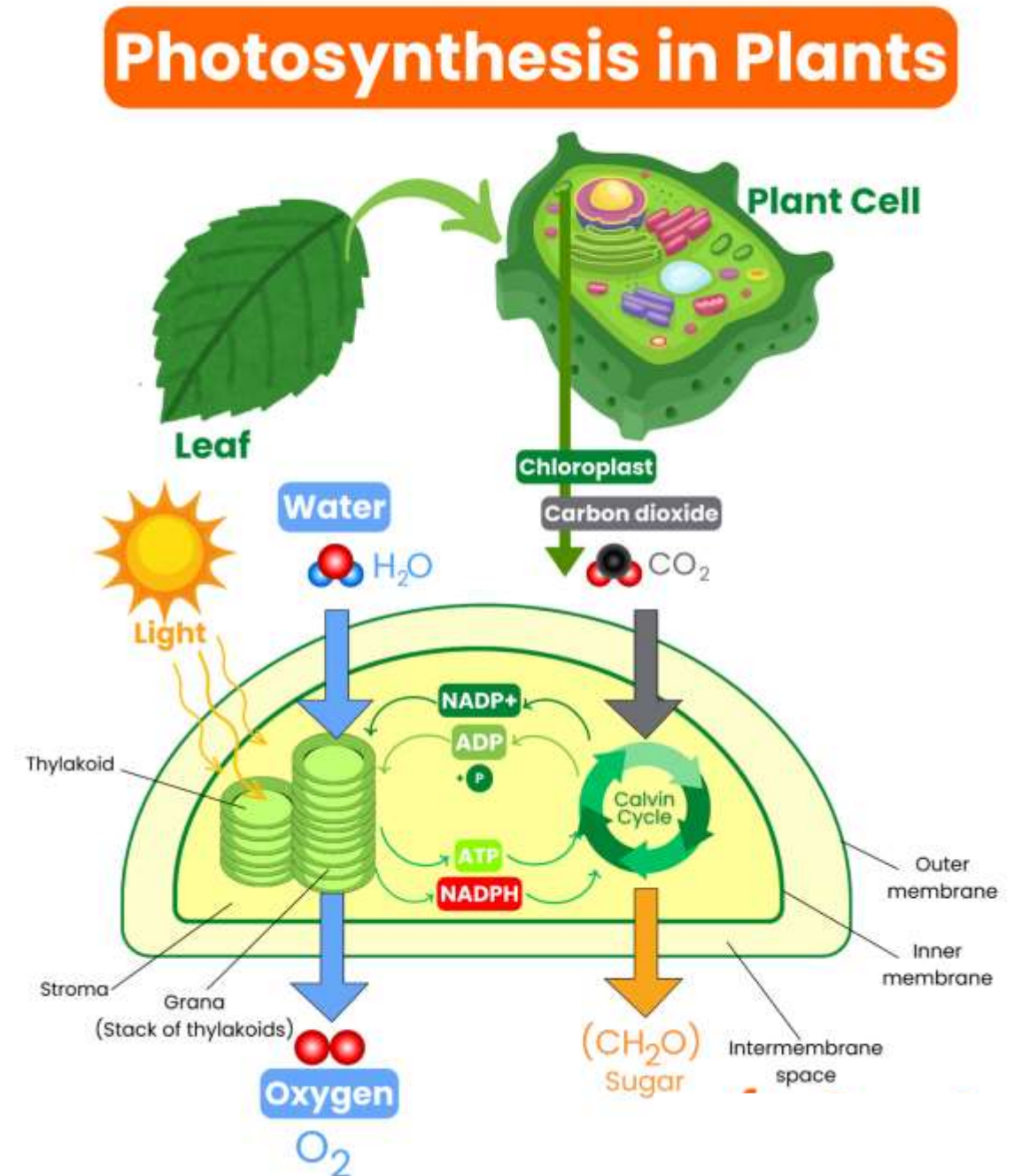
Photosynthesis

Key Steps:

- 1. Light-dependent reactions:** Occur in the thylakoid membranes of chloroplasts. Light energy is absorbed by chlorophyll, splitting water molecules and releasing oxygen. **ATP** and **NADPH** are produced.
- 2. Calvin Cycle (Light-independent reactions):** Occurs in the stroma of chloroplasts. **ATP** and **NADPH** are used to convert carbon dioxide into glucose.

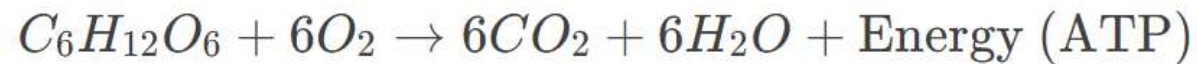
Importance:

- Produces oxygen, which is essential for most living organisms.
- Converts solar energy into a usable form of energy (glucose).
- Forms the basis of the food chain.



Cellular Respiration

- Cellular respiration is the process by which cells break down glucose to release energy in the form of ATP (adenosine triphosphate).
- It occurs in both plants and animals.
- Energy produced is essential for growth, movement, repair, and other cellular activities.

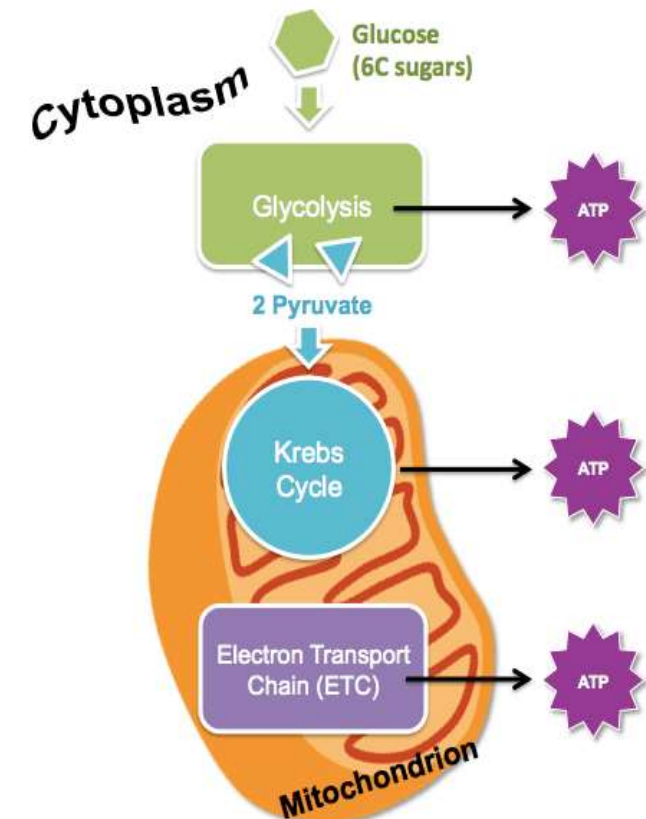


Key Steps:

1. Glycolysis: Occurs in the cytoplasm. Glucose is broken down into two molecules of pyruvate, producing 2 ATP and NADH.

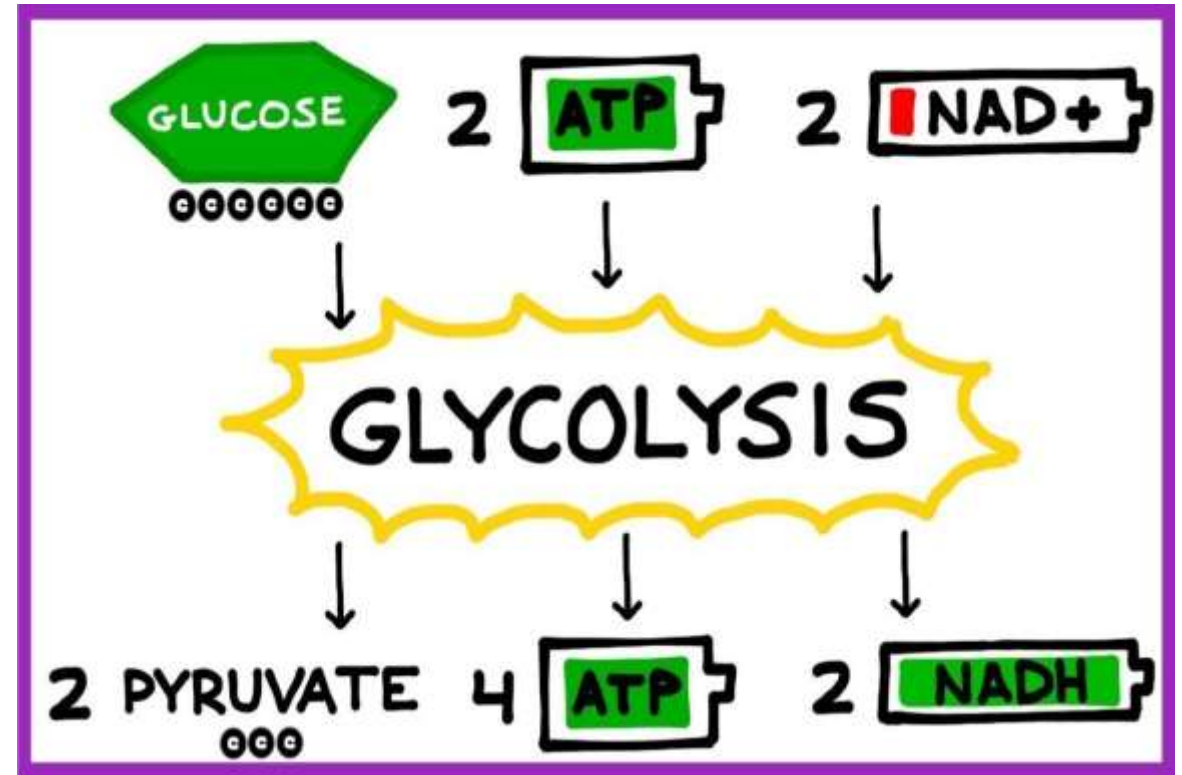
2. Krebs Cycle (Citric Acid Cycle): Occurs in the mitochondrial matrix. Pyruvate is further broken down, releasing carbon dioxide and producing ATP, NADH, and $FADH_2$.

3. Electron Transport Chain (ETC): Occurs in the inner mitochondrial membrane. NADH and $FADH_2$ donate



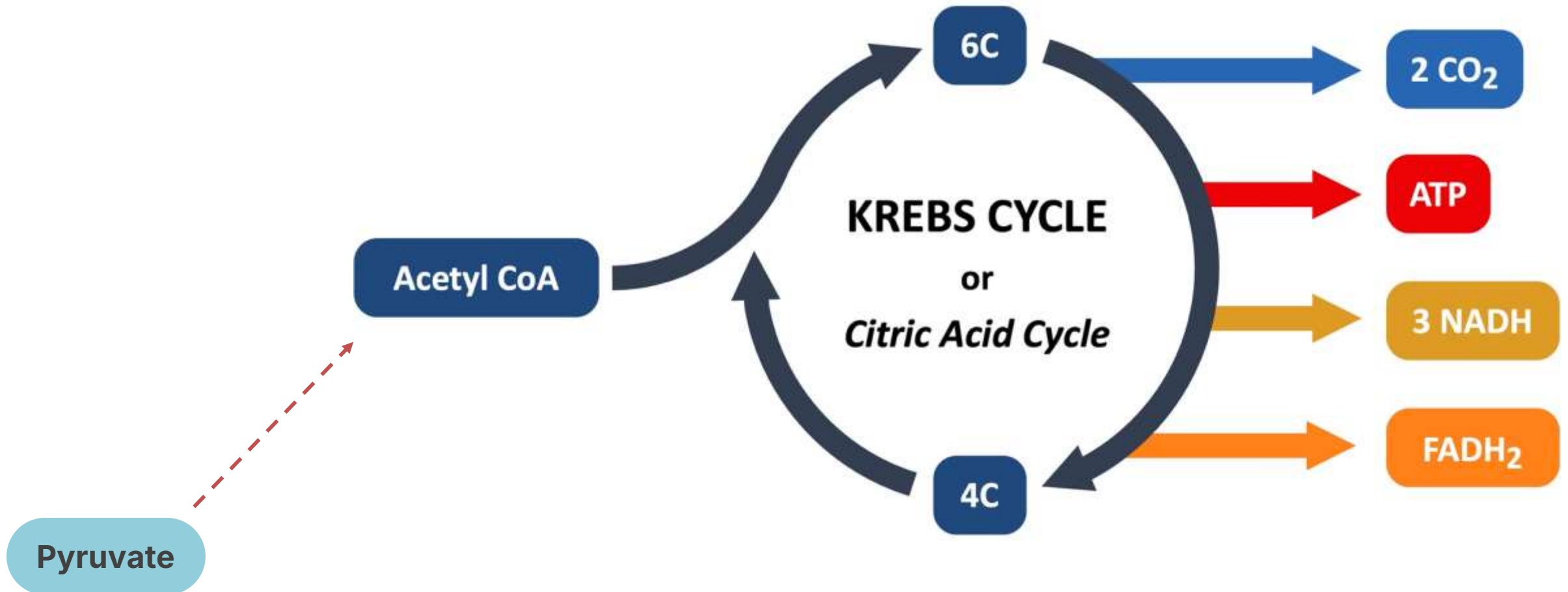
Key Steps

1. Glycolysis: Occurs in the cytoplasm. Glucose is broken down into two molecules of pyruvate, producing 2 ATP and NADH.



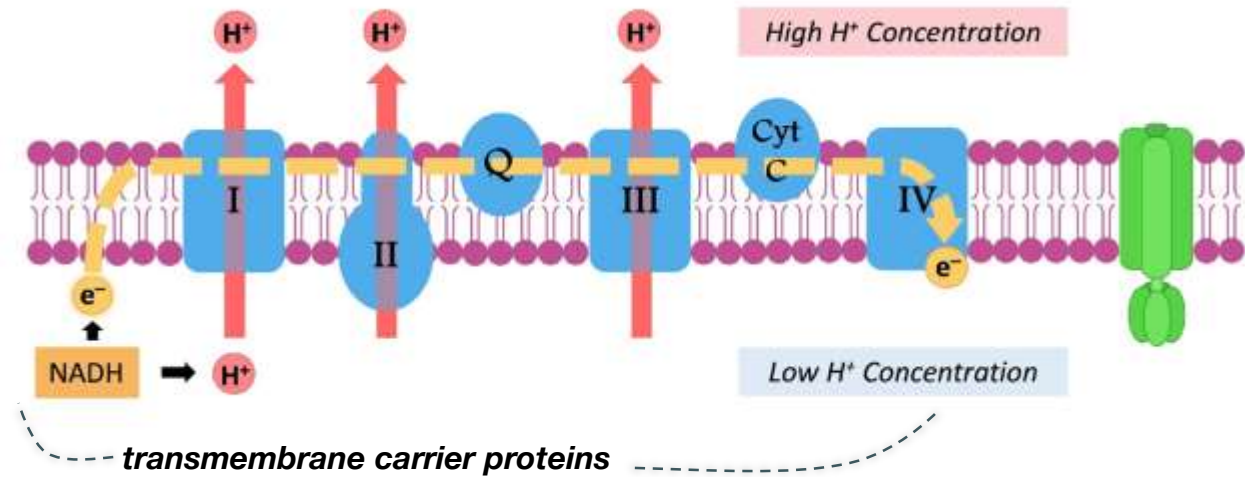
Key Steps

2.Krebs Cycle (Citric Acid Cycle): Occurs in the mitochondrial matrix. Pyruvate is further broken down, releasing carbon dioxide and producing ATP, NADH, and FADH₂.



Key Steps

3. Electron Transport Chain (ETC): Occurs in the inner mitochondrial membrane. NADH and FADH_2 donate electrons, driving the production of ATP. Oxygen is the final electron acceptor, forming water.

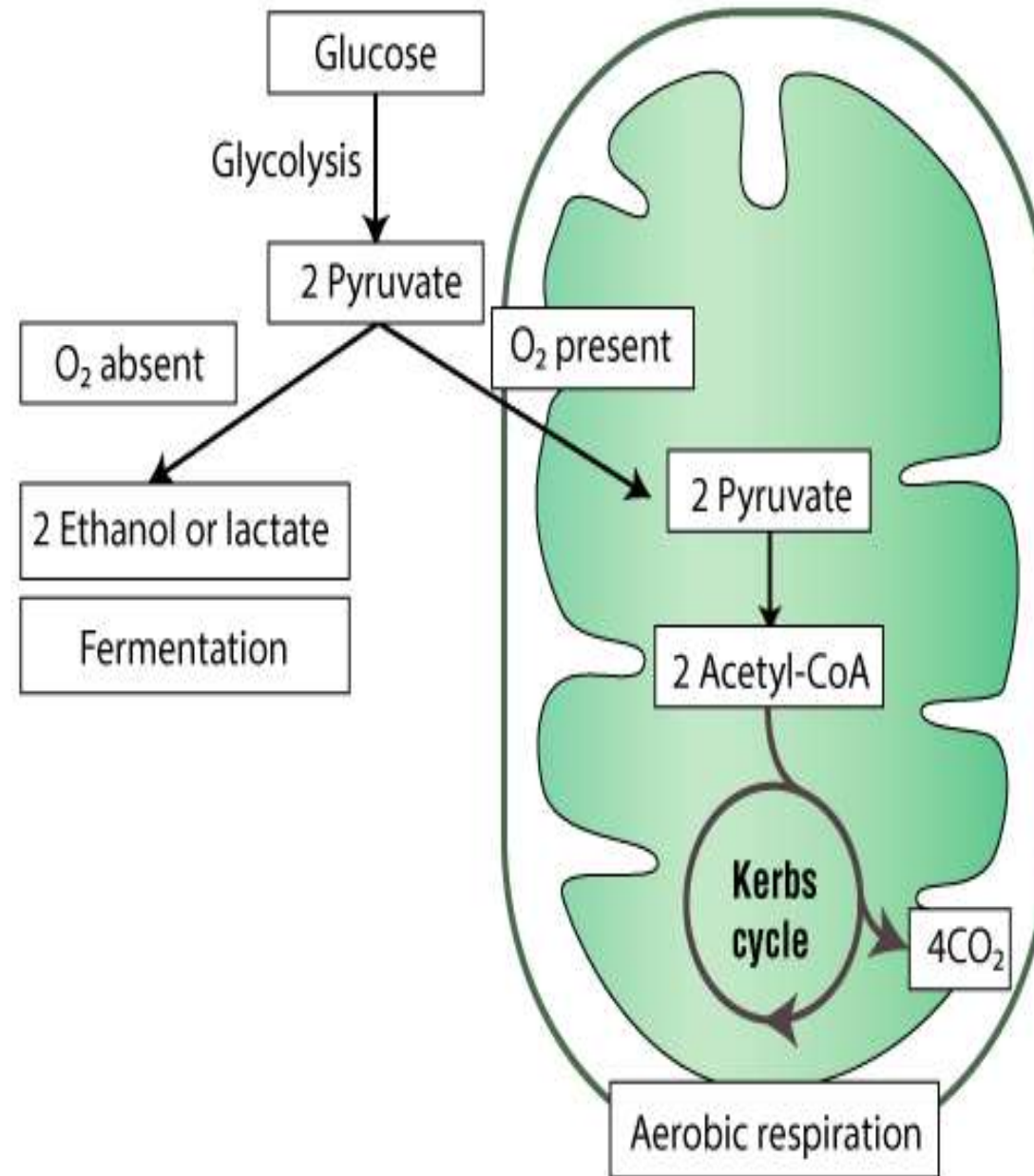


Pathway	Substrate-Level Phosphorylation	Oxidative Phosphorylation	Total ATP
Glycolysis	2 ATP	2 NADH = 4 	6 
CoA		2 NADH = 6 ATP	6
Krebs Cycle	2 ATP	6 NADH = 18 ATP 2 FADH_2 = 4 ATP	24
TOTAL	4 ATP	32 ATP	36 

Types of Cellular Respiration

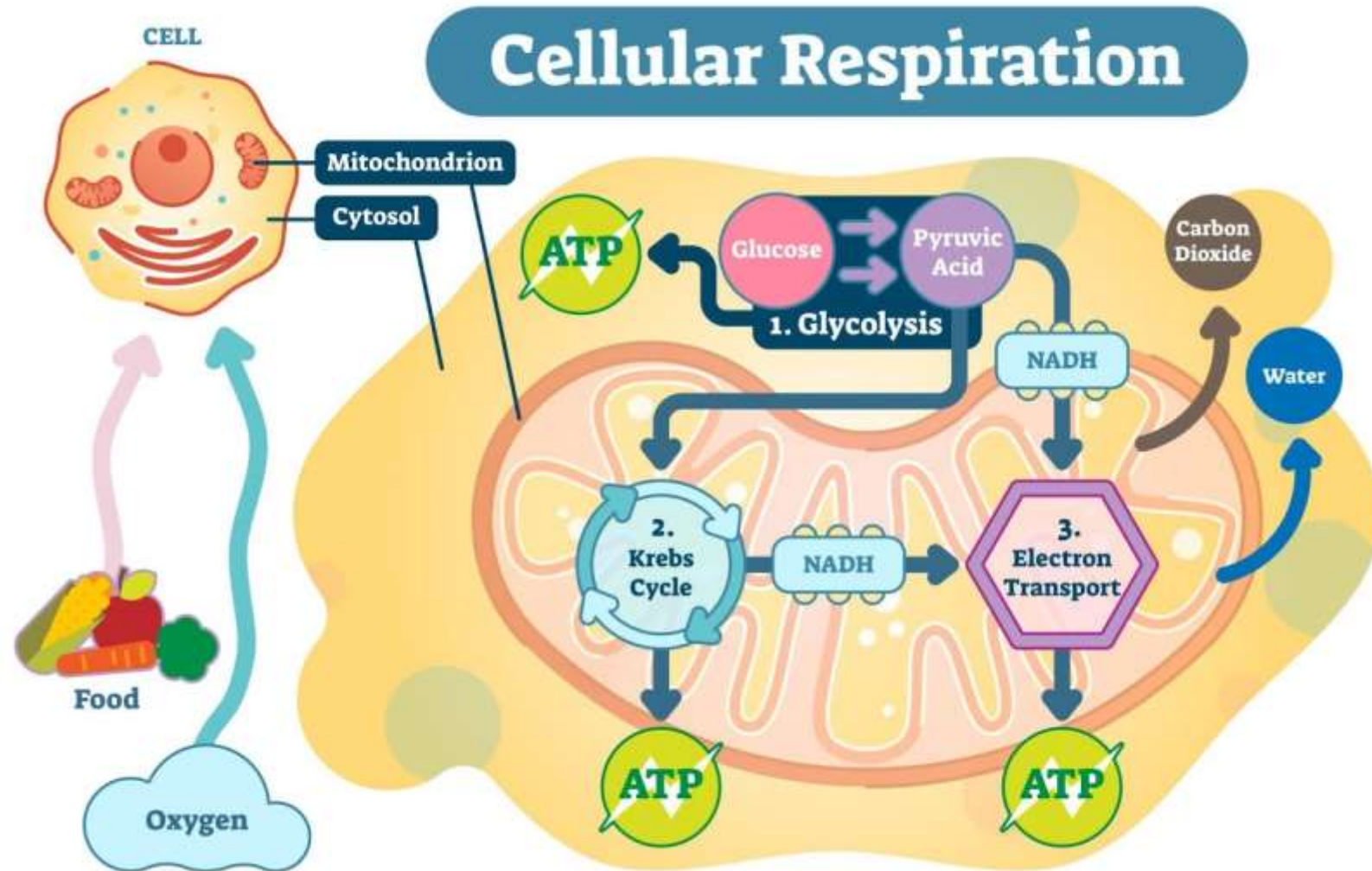
Type	Description	Oxygen Requirement
Aerobic Respiration	Glucose is completely broken down into carbon dioxide and water.	Requires oxygen
Anaerobic Respiration	Glucose is partially broken down, producing less energy.	Occurs without oxygen

Types of Cellular Respiration



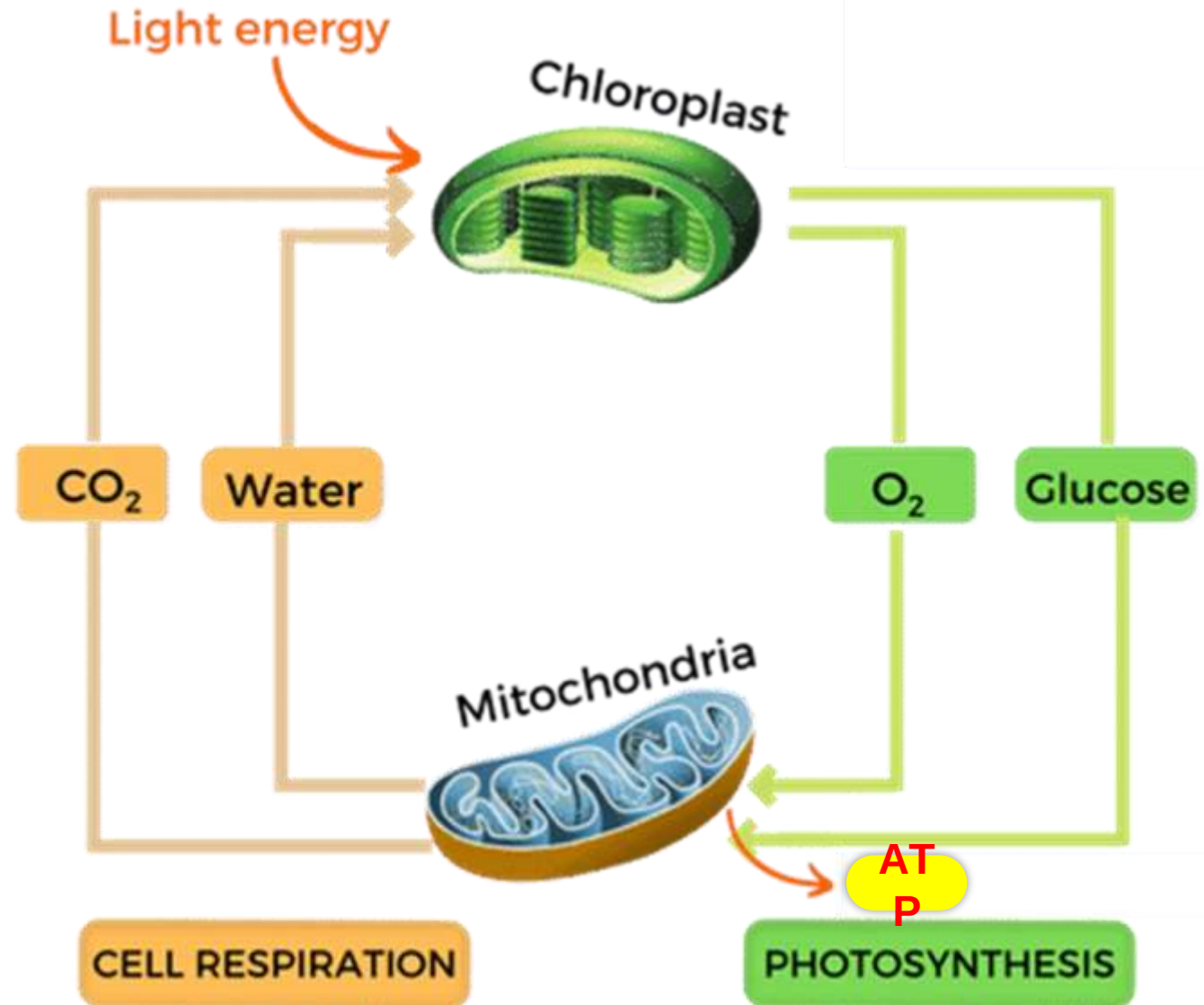
Cellular Respiration in **Animals**

- Occurs in the **mitochondria** of animal cells.
- Animals obtain glucose from food.
- Oxygen is inhaled through breathing.
- End products: **CO₂**, **H₂O**, and **ATP (energy)**.
- Excess CO₂ is exhaled.

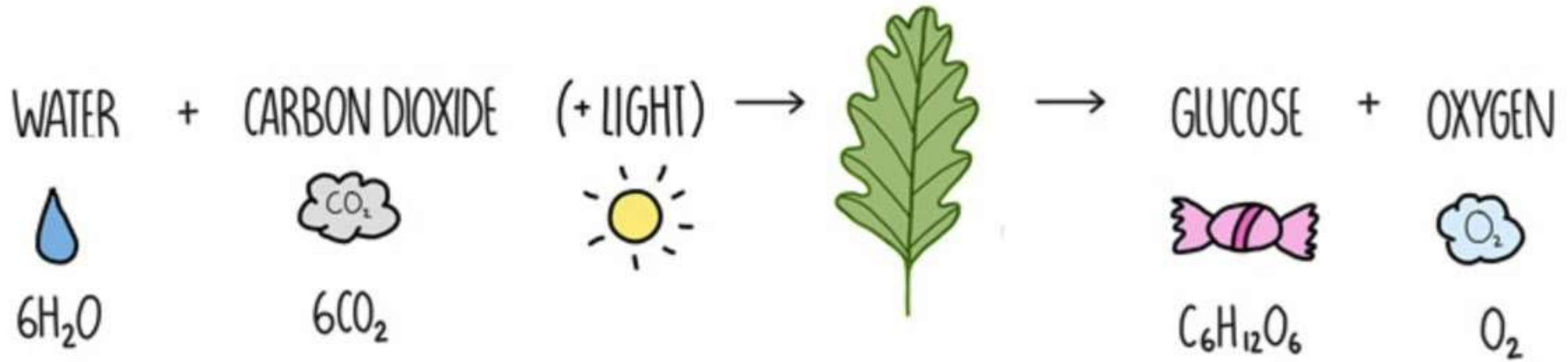


Cellular Respiration in **Plants**

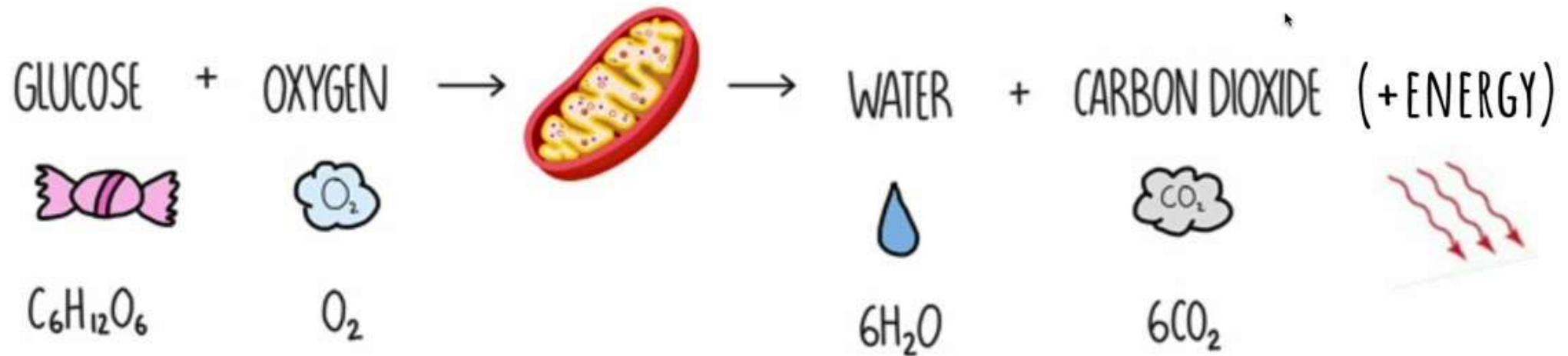
- Occurs in **all parts of the plant**, especially in mitochondria of cells.
- Glucose is produced during **photosynthesis** in chloroplasts and then used for respiration.
- Plants also take in oxygen and release CO_2 , especially at night when photosynthesis stops.
- Respiration is essential for **root, stem, and leaf functions**.



Photosynthesis



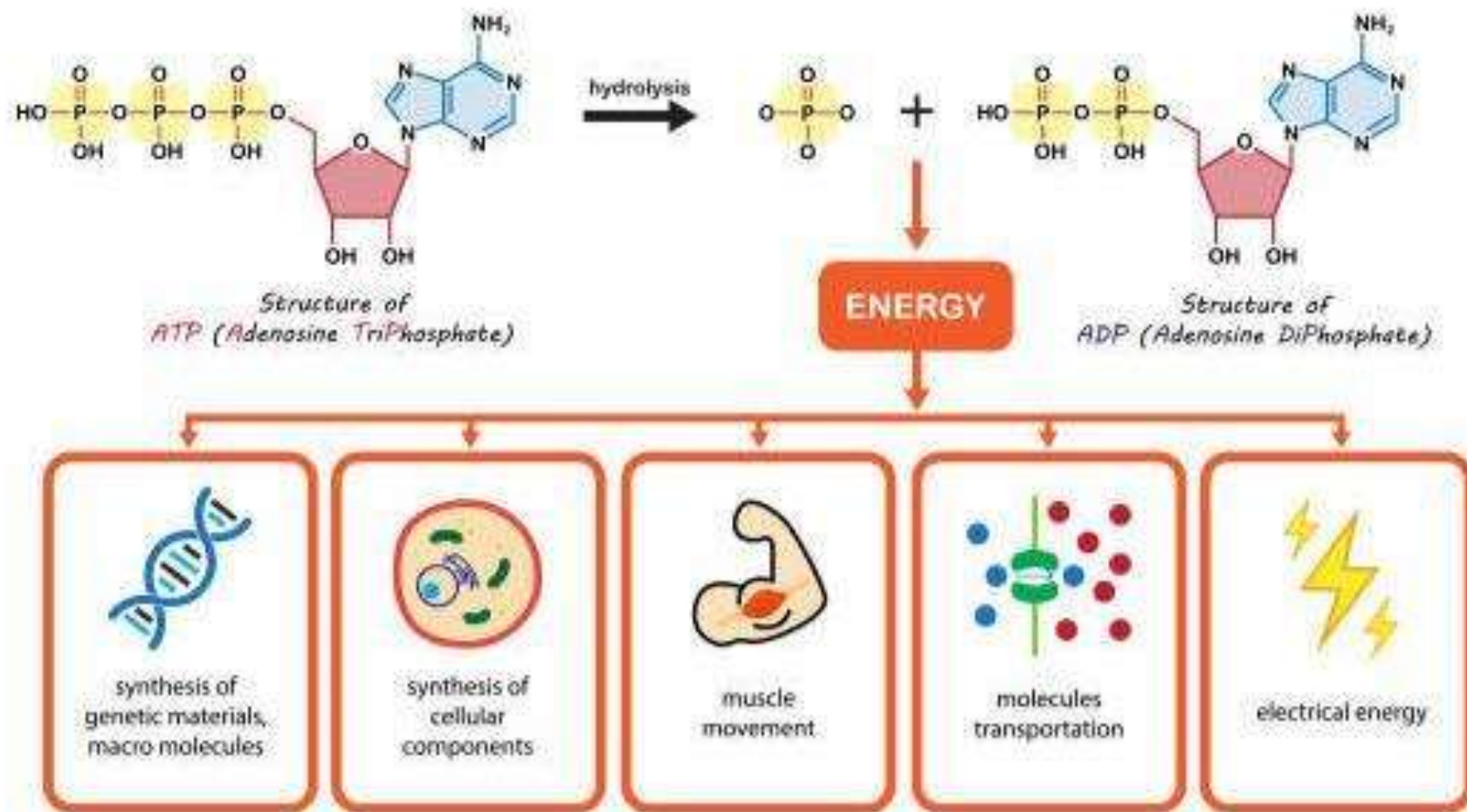
Cellular Respiration



Comparison Between photosynthesis and respiration

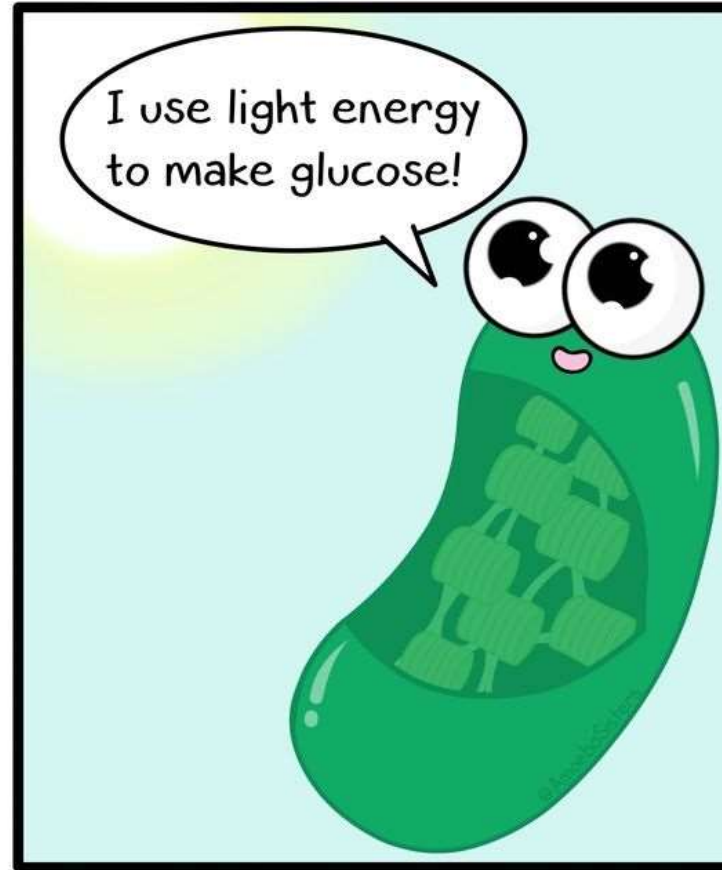
	Respiration	Photosynthesis
Inputs	Oxygen & glucose	Carbon dioxide, light, water
Products	Carbon dioxide, water & energy (ATP)	Oxygen, glucose & water
Type of cells	All cells	Plants & some protists & bacteria
Location in cells	Cytoplasm (anaerobic) & mitochondria (aerobic)	Chloroplast
Where energy comes from	Carbon compounds (chemical breakdown)	Light
Role of ATP	End product	Carries energy of light reactions into dark reactions
Names of reactions	Glycolysis, fermentation, Krebs cycle	Light reactions – Photosystems I & II & electron transport chain Dark reactions – Calvin cycle

What is **ATP** used for?

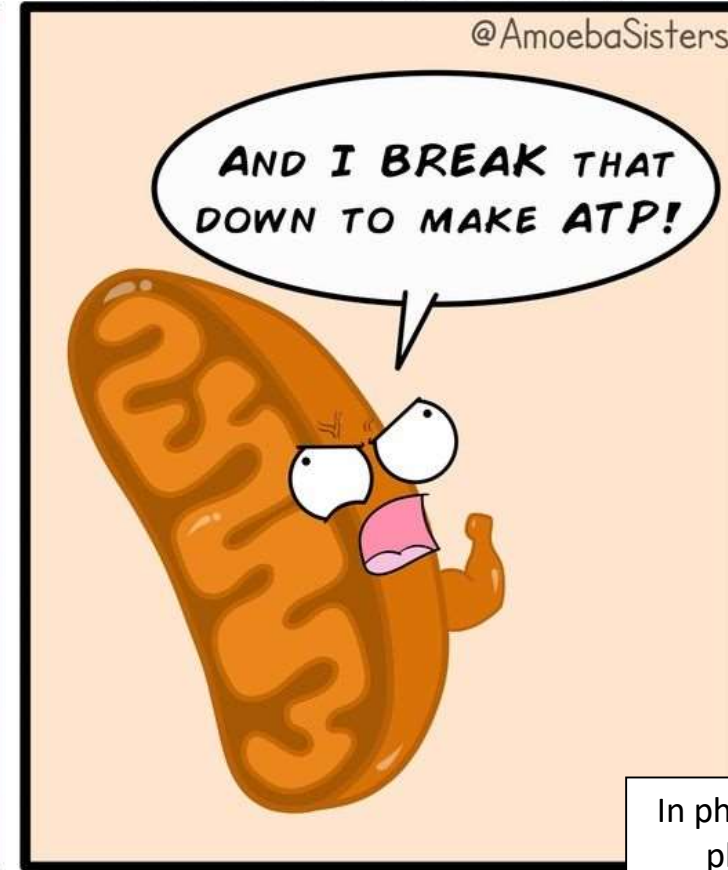


Thank You!

CHLOROPLAST VS MITOCHONDRION



- Found in (most) photosynthetic eukaryotes
- Used for photosynthesis



- Found in (most) eukaryotes
- Used for cellular respiration

In photosynthetic plants too!